

Answer to the question no: 1

Software Quality Assurance: Software Quality Assurance is a systematic process to ensure the software was developed and maintained according to quality standards, process and requirements.

Software often used various critical areas like hospital, banking sector, security, government services etc. If software doesn't have any standard quality, it'll cause privacy breaches, data leaks, user dissatisfaction. If it's happening, the brand of that software will face reputation and ethical issue. So, it's important to maintain software quality.

Answer to the question no: 2

Quality Assurance (QA)	Quality Control (QC)
1. QA is Process-oriented.	1. QC is Product-oriented.
2. QA focuses on preventing defects.	2. QC focuses on identifying defects.
3. It ensures the development process follows standards and best practices.	3. It verifies that the final product meets quality requirements.

Quality Control (QC)	Testing
1. A broad process for verifying software quality.	1. A specific activity within QC.
2. Includes inspections, reviews, walkthroughs, and testing.	2. Involves executing software to detect defects.
3. May use both static and dynamic techniques.	3. Primarily a dynamic technique (running the software).

Quality Assurance (QA)	Testing
1. Focuses on improving the development process.	3. Focuses on evaluating the software product.
2. Prevents defects before they occur.	2. Finds defects after or during development.
3. Process-oriented.	3. Product-oriented.

Answer to the question no: 3

Artificial Intelligence (AI) is transforming software testing by making testing faster, smarter, and more efficient. Instead of relying entirely on manual effort, AI automates many testing activities, predicts potential defects, and helps improve software quality. Below I'm discussing this:

1. **Automated Test Case Generation:** AI can automatically generate test cases from requirements, user stories, or application behaviour, reducing manual effort.
2. **Self-Healing Test Scripts:** AI-powered tools can automatically update test scripts when the application's UI changes, reducing test maintenance.
3. **Smarter Test Execution:** AI prioritizes the most critical test cases based on code changes, risk, and historical defects, saving time.
4. **Defect Prediction:** AI analyses historical project data to predict which modules are most likely to contain defects, allowing teams to focus testing efforts.
5. **Faster Bug Detection:** Machine learning algorithms identify patterns in logs and test results to detect bugs more quickly and accurately.
6. **Test Data Generation:** AI creates realistic and diverse test data automatically, improving test coverage.
7. **Improved Regression Testing:** AI selects only the most relevant regression tests after code changes, reducing execution time.
8. **Natural Language Processing (NLP):** AI can convert requirements written in plain English into automated test cases or testing scenarios.
9. **Continuous Testing in DevOps:** AI supports Continuous Integration and Continuous Deployment (CI/CD) by enabling intelligent, automated testing throughout the software development lifecycle.

Answer to the question no: 4

Responsibilities and Daily Activities of Different Software Quality Roles:

1. QA Engineer (Quality Assurance Engineer): A QA Engineer focuses on entire software development process follows quality standards and best practices. Their goal is to prevent defects before they occur.

Key responsibilities include:

- Defining and improving QA processes.
- Reviewing software requirements and specifications.
- Creating test plans and test strategies.

Daily Activities

- Attend daily Scrum/stand-up meetings.
- Review requirements and user stories.
- Prepare test plans and testing strategies.
- Review test cases written by the testing team.
- Participate in requirement and design review meetings.

2. Software Test Engineer: A Software Test Engineer focuses on finding defects in the software by performing different types of testing. Their goal is to verify that the software meets functional and non-functional requirements.

Key responsibilities include:

- Designing and executing test cases.
- Performing manual and automated testing.
- Reporting and tracking software defects.
- Verifying bug fixes.
- Performing regression testing.
- Ensuring software meets customer requirements.
- Maintaining test documentation.

Daily Activities

- Analyse requirements and acceptance criteria.
- Write and update test cases.
- Execute manual or automated tests.
- Report bugs using tools like Jira or Trello.
- Re-test fixed defects.
- Perform regression and smoke testing.
- Prepare daily testing status reports.
- Work closely with developers to resolve issues.

3. AI Quality Engineer: An AI Quality Engineer ensures the quality, reliability, fairness, and performance of AI and Machine Learning systems. Unlike traditional software testing, this role also evaluates AI models, data quality, and ethical considerations.

Key responsibilities include:

- Test AI and machine learning models.
- Validate training and testing datasets.
- Measure model accuracy, precision, recall, and other performance metrics.
- Verify AI outputs against expected behaviour.
- Monitor AI systems after deployment.

Daily Activities

- Validate datasets for completeness and quality.
- Test AI model predictions.
- Analyse false positives and false negatives.
- Measure model performance metrics.
- Run automated AI evaluation pipelines.
- Monitor production AI systems.
- Collaborate with data scientists and machine learning engineers.

Answer to the question no: 5

